ADJOINING A ROOT OF A POLYNOMIAL IN THE DERIVATIVE IMAGE STAYS REGULAR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Regularity ascends to polynomial rings. Algebra, Lemma \ref{algebra-lemma-regular-goes-up}, cited at proof lines 11852-11853. Makes the polynomial ring regular.
- Jacobian criterion for a quotient by an element whose derivative is a unit along the quotient. Lemma \ref{lemma-quotient-regular}, cited at proof lines 11853-11855. Gives regularity of the quotient.

Source reference. Original lemma 07PG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9796–9796. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07NF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9760–9760. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07PF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9795–9795. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a regular ring. Let $f \in R$. Assume there exists a derivation $D : R \to R$ such that $D(f)$ is a unit of R. Then $R[z]/(z^n - f)$ is regular for any integer $n \geq 1$. More generally, $R[z]/(p(z) - f)$ is regular for any $p \in \mathbf{Z}[z]$.*

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Proof. By Algebra, Lemma 07NF we see that $R[z]$ is a regular ring. Apply Lemma 07PF to the extension of D to $R[z]$ which maps z to zero. This works because D annihilates any polynomial with integer coefficients and sends f to a unit. $\square$